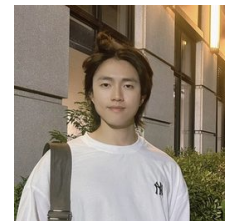


Hao-Chun Liang

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EDUCATION

National Yang Ming Chiao Tung University <i>M.Eng. in IPSI (Chip Design and Automation Group) GPA: 4.3/4.3</i>	Sep. 2025 – Jun. 2027 (Exp.) <i>Hsinchu, Taiwan</i>
National Tsing Hua University <i>B.S. in Electrical Engineering GPA: 4.1/4.3 Distinction Graduation</i> Transferred from Dept. of Physics (Top 1 of 65; College of Science Elite Student Award).	Sep. 2020 – Jun. 2024 <i>Hsinchu, Taiwan</i>

Graduate Coursework (NYCU): ICLAB (A+, 5 of 143 passed the on-site test), DSP-VLSI (A+), Deep Learning (A+), Stochastic Processes (A+)

Undergraduate Coursework (NTHU): ICLAB (A+), VLSI Testing (A+), VLSI (A), Computer Architecture (A-), Machine Learning (A), Communication System (A+)

WORK EXPERIENCE

Digital IC Design Intern <i>Andes Technology Corporation Compute Acceleration R&D Division Deep Learning Accelerator</i>	Jul. 2026 – Present <i>Hsinchu, Taiwan</i>
Graduate Research Assistant <i>National Yang Ming Chiao Tung University Parallel Computing System Laboratory</i>	Sep. 2025 – Jun. 2027 (Exp.) <i>Hsinchu, Taiwan</i>
IT Manager <i>National Yang Ming Chiao Tung University Parallel Computing System Laboratory</i>	Sep. 2025 – Jun. 2027 (Exp.) <i>Hsinchu, Taiwan</i>
Digital IC Design Intern <i>Andes Technology Corporation Compute Acceleration R&D Division Deep Learning Accelerator</i>	Nov. 2024 – Aug. 2025 <i>Hsinchu, Taiwan</i>

AWARDS & CERTIFICATES

Elite New Graduate Scholarship <i>Industry Academia Innovation School, NYCU</i>	Sep. 2025 <i>Hsinchu, Taiwan</i>
Excellent Poster Award <i>Taiwan and Japan Conference on Circuits and Systems (TJCAS)</i>	Aug. 2025 <i>Kaohsiung, Taiwan</i>
Cell-Based Digital Chip Design and Implementation (ADFP - TSMC 16nm) <i>Taiwan Semiconductor Research Institute (TSRI)</i>	Aug. 2024 <i>Hsinchu, Taiwan</i>
Distinction Graduation (Top 5% of 116 in Professional-Course GPA) <i>National Tsing Hua University Electrical Engineering</i>	May 2024 <i>Hsinchu, Taiwan</i>
College of Science Elite Student Award (Top 1 of 65) <i>National Tsing Hua University College of Science</i>	Apr. 2021 <i>Hsinchu, Taiwan</i>
Academic Excellence Award (Received 3 times) <i>National Tsing Hua University</i>	2020 – 2022 <i>Hsinchu, Taiwan</i>

PUBLICATIONS

UniTrac: A Unified FPGA Accelerator for GNN-based Particle Track Reconstruction in HEPs <i>ACM Transactions on Reconfigurable Technology and Systems (2nd-Major-Revision) Second + Corresponding Author</i>
Real-Time GPU Kalman-Filter Tracking via Kernel Refactoring and INT8 Surrogates <i>Fast Machine Learning for Science Conference 2025, ETH Zurich First Author</i>
An Integrated FPGA Implementation of Complete GNN-Based Trajectory Reconstruction <i>Taiwan and Japan Conference on Circuits and Systems (TJCAS) 2025 Second Author</i>

A High-Performance Implementation of GNN-Based Trajectory Reconstruction on FPGA

VLSI Design/CAD Symposium 2025 | Second Author

Fine-Grained Image Recognition from Scratch with Teacher-Guided Data Augmentation

arXiv:2507.12157 | Fifth Author

PROJECTS

L/SDMA Untimed SystemC Algorithm/TLM-2 Model <i>Andes Technology Corporation</i>	Jul. 2026 – Present <i>Hsinchu, Taiwan</i>
AI Agent Automation Flow <i>Andes Technology Corporation</i>	Jul. 2026 – Present <i>Hsinchu, Taiwan</i>
Regfile RTL/C Code Generation IR Optimization <i>Andes Technology Corporation</i>	Jul. 2026 – Present <i>Hsinchu, Taiwan</i>
Trajectory Reconstruction Accelerator TSMC 28nm Tapeout <i>National Yang Ming Chiao Tung University Parallel Computing System Laboratory</i>	Jul. 2026 – Present <i>Hsinchu, Taiwan</i>
Regfile RTL/C Code Generation from Scratch <i>Andes Technology Corporation</i>	Nov. 2024 – Aug. 2025 <i>Hsinchu, Taiwan</i>
Ultra-Lightweight HDC-CNN for NeurIPS 2025 Weak Lensing Challenge <i>NeurIPS FAIR Universe Challenge</i>	Oct. 2025 – Nov. 2025 <i>Top-20 Ranking</i>
NVIDIA NSYS/NCU Profile and Analysis of Token Reduction for ViT <i>National Yang Ming Chiao Tung University Parallel Computing System Laboratory</i>	Oct. 2025 – Nov. 2025 <i>Hsinchu, Taiwan</i>
RISC-V 5-Stage Pipelined CPU Core <i>National Tsing Hua University</i>	Dec. 2023 – Jan. 2024 <i>Hsinchu, Taiwan</i>

SKILLS

Programming Languages: C/C++, SystemVerilog, Python, CUDA, Makefile

Tools: VCS, DC, Vivado, PrimeTime, Spyglass, JasperGold, Innovus, Calibre, NVIDIA NSYS/NCU, Git

Domains: Chip Design, Domain-Specific Architecture, GPU Performance Engineering, AI Agent Automation